



# DELYAN BOYCHEV

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Sofia, Bulgaria

## ABOUT ME

Informatics undergraduate and INSAIT research scholar. I work on multimodal learning and natural language processing, and more recently on efficient model training and inference.

## EDUCATION

**INSAIT Scholar (highly selective 4-year program)**

Sofia, Bulgaria

**Faculty of Mathematics and Informatics, Sofia University**

*Bachelor of Informatics*

*Sept. 2024 – Expected 2028*

## EXPERIENCE

**IOAI Lecturer | Bulgarian IOAI Team**

June 2025 – Present

- Lecturing and mentoring students preparing for the International Olympiad in Artificial Intelligence (IOAI).

**Undergrad Research Scholar | INSAIT**

Oct. 2024 – Present

- Researching vision-language models for remote sensing and 3D scene understanding under the supervision of Dr. Danda Paudel.

**NLP Research Intern | Graphwise**

July 2026 – Sept. 2026

- Built NLP pipelines for table entity reconciliation, knowledge-base linking, and structured extraction from unstructured text.
- Optimized language-model pipelines for deployment, improving inference cost and latency.

**Quant ML Intern | Man Group**

July 2025 – Sept. 2025

- Conducted Quant ML research, focusing on developing and evaluating machine-learning models for systematic trading and signal forecasting.

## PUBLICATIONS

**OSMDA: OpenStreetMap-based Domain Adaptation for Remote Sensing VLMs** | *arXiv Preprint, 2026*

- Stefan Maria Ailuro, Mario Markov, Mohammad Mahdi, **Delyan Boychev**, Luc Van Gool, Danda Pani Paudel
- Self-contained domain adaptation using OpenStreetMap supervision without manual labels or a stronger external teacher; under review.
- **Paper:** [arxiv.org/abs/2603.11804](https://arxiv.org/abs/2603.11804)

**ImagiNet: A Multi-Content Benchmark for Synthetic Image Detection** | *AAAI 2025 Workshop Paper*

- **Delyan Boychev**, Radostin Cholakov — AAAI 2025 Workshop on Datasets and Evaluators of AI Safety
- A balanced 200K-image benchmark spanning four content types and multiple open and proprietary generators, with strong contrastive-learning baselines.
- **Paper:** [arxiv.org/abs/2407.20020](https://arxiv.org/abs/2407.20020)

**Interpretable Computer Vision Models through Adversarial Training** | *arXiv Preprint, 2023*

- Single-author empirical study of how adversarial training changes learned representations and model interpretability.
- **Paper:** [arxiv.org/abs/2307.02500](https://arxiv.org/abs/2307.02500)

## PROJECTS

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**DisentangledFlash** | *Triton Attention Kernel* Aug. 2026 – Present

- Fuses DeBERTa-v2/v3 attention into a streaming Triton kernel without materializing the full attention matrix; broader-family extensions are underway, with publication under consideration.
- Benchmarked at up to  $5.18\times$  faster forward and  $2.15\times$  faster forward+backward than Hugging Face Transformers eager.
- **Code:** [github.com/delyan-boychev/disentangled-flash](https://github.com/delyan-boychev/disentangled-flash) | **PyPI:** [pypi.org/project/disentangled-flash](https://pypi.org/project/disentangled-flash)

**LavBench** | *ML Competition Platform* June 2026 – Present

- A secure, bilingual platform for machine-learning competitions, built for Bulgarian AI Olympiad selection and national contests.
- Runs notebook and Python submissions in hardened Docker sandboxes; provides live leaderboards, double-blind jury review, custom evaluators, and distributed workers.
- **Code:** [github.com/delyan-boychev/lavbench](https://github.com/delyan-boychev/lavbench) | **Docs:** [lavbench.readthedocs.io](https://lavbench.readthedocs.io)

**Generative Translation** | *Coursework in Information Search and Retrieval: Application of Deep Learning* Feb. 2026

- Built a decoder-only PyTorch Transformer for Bulgarian-to-English translation with custom BPE, masked attention, KV caching, and beam search.
- Achieved 42.57 BLEU-4 with 12.26 test perplexity using beam search.
- **Code:** [github.com/delyan-boychev/gen-trans](https://github.com/delyan-boychev/gen-trans)

**Vision-Language Models for Satellite Imagery Understanding** | *Research at INSAIT* Dec. 2025 – Present

*Advised by Dr. Danda Paudel*

- Adapting vision-language models to satellite imagery using scalable supervision derived from OpenStreetMap.

**Spatial Modeling for Point-Cloud Understanding** | *Research at INSAIT* June 2025 – Present

*Advised by Dr. Danda Paudel*

- Aligning 3D point-cloud and Gaussian representations for structured spatial reasoning and scene understanding.

**C++ Coursework** | *DSA and OOP Course Projects* 2025

- Built a dependency-aware expression solver with polymorphic tokenization and two-stack evaluation, and a modular digital audio workstation with timeline editing, effects, and audio I/O.
- **Solver:** [partial-solver](#) | **Audio Workstation:** [dig-audio-workstation](#)

**ImagiNet** | *Synthetic Image Dataset* 2024

- A 144 GB, 200K-image dataset for synthetic-image detection, balanced across photos, paintings, faces, and uncategorized imagery.
- Supports both real-versus-synthetic classification and generator identification across open and proprietary image generators.
- **Dataset:** [huggingface.co/datasets/delyanboychev/imagenet](https://huggingface.co/datasets/delyanboychev/imagenet)

**Gradient Cache Contrastive Learning** | *PyTorch Package* 2023 – 2024

- Implemented gradient caching in PyTorch to scale contrastive learning beyond GPU memory limits.
- Reproduces large-batch gradients from smaller chunks; supports SimCLR, SupCon, SelfCon, mixed precision, and distributed training.
- **Code:** [github.com/delyan-boychev/grad-cache-con-learning](https://github.com/delyan-boychev/grad-cache-con-learning) | **PyPI:** [pypi.org/project/grad-cache-con-learning](https://pypi.org/project/grad-cache-con-learning)

## HONORS

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- Scholarship Recipient, EXPLORER Program, INSAIT (2024)
- European Space Agency Award from EUCYS'24, Katowice, Poland
- International Olympiad in Artificial Intelligence 2024, Team Bronze Medal
- Regeneron ISEF'23 Finalist, Dallas, Texas
- John Atanasoff Presidential Award in the category “Debut breakthrough in computer technology” (2023)

## TECHNICAL SKILLS

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- **Languages:** Python, C++, TypeScript, JavaScript
- **ML/Research:** PyTorch, Triton, vLLM, Hugging Face Transformers
- **Data Science:** NumPy, pandas, scikit-learn
- **Web & Backend:** React, Vite, Tailwind CSS, Flask, Celery, PostgreSQL
- **Tools:** Linux, Docker, Google Cloud, CI/CD, Git

## LANGUAGES

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- Bulgarian — Native
- English — IELTS Academic, 7.0/9.0